Jose Torres

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EDUCATION

Brigham Young University Provo, UT Expected Apr 2028

B.S. Computer Engineering, *Minor in Computer Science*

Relevant Coursework: Digital Systems, Embedded Programming, Circuits; in progress: Embedded Linux Systems, Signals & Systems, Electronic Circuits

EXPERIENCE

CAST Engineering Solutions LLC General contractor: remodels, electrical and low-voltage work, Lehi, UT

Project Manager & Estimator Apr 2026 – Present

- Built the business systems CAST lacked at my family's general contracting company, which ran on one person who bid every job, worked alongside the crew, paid workers and reviewed revenue once a year: created estimating, sales, finance and compliance processes, with reusable Excel templates for labor and materials takeoffs
- Run bids, estimates, sales, materials ordering, marketing and lead generation, and license, insurance and payroll paperwork; coordinate a two-person field crew while studying full time
- Estimate electrical and low-voltage work, including CCTV and network cabling, from architectural drawings: device takeoffs (e.g., 15 cameras, 16 data drops, 12 wireless access points on a retail buildout), labor hours and scope exclusions

Technical Assistant Oct 2024 – Jun 2025

- Assisted with electrical and low-voltage installs, including IP surveillance (Bosch cameras, 5–7 per site), smart locks, door sensors and alarm integration, and network configuration

Sales: Premium Wireless (Jun 2025 – Apr 2026), Top 10 in the Utah/Colorado region; Apollo Home Power (Sep 2024 – Apr 2025), sold 20+ residential energy storage systems; Sunrun Solar (Apr 2024 – Aug 2024), door-to-door solar sales

Teaching: Language Teacher & Tutor, Missionary Training Center (Aug 2022 – Apr 2024); trained and mentored 100+ people in Spanish

ENGINEERING PROJECTS

Laser Tag Optical Board ngspice, KiCad, MCP6001 op amps

- Designed a 3.3 V photodiode receiver with two op amps (transimpedance stage plus two inverting bandpass stages, ~21,000 V/V gain, 500 Hz–15 kHz passband) that recovers a 40 nA signal from 1 mA ambient light and 400 nA optical noise
- Simulated 2.2 V of signal over noise (spec 0.2 V) with flat response across nine player frequencies; breadboard test against a mounted laser tag gun measured 0.26 V over noise (spec 0.033 V) from 100 kS/s oscilloscope captures
- Built a CAT4104 LED-driver transmitter and designed the two-sided KiCad PCB integrating both circuits; verified the assembled board

Embedded Linux Space Invaders C/C++, SystemVerilog, Vivado, Xilinx PYNQ-Z2

- Built a 60 fps interrupt-driven game over the HDMI framebuffer (team of three), meeting limits of under 500 system calls and 3 ms per sprite
- Wrote a Linux kernel module for an audio codec (interrupt handler keeping FIFOs full at 48 kS/s, ioctl looping) and a sysfs driver for a custom timer I designed in SystemVerilog as an AXI peripheral and verified with AXI VIP simulation
- Implemented a scatter-gather DMA driver (AMD AXI CDMA) to offload sprite copies, plus UIO user-space drivers for GPIO and the interrupt controller

FPGA Digital Design Basys3, SystemVerilog, Vivado

- Built a 640×480 60 Hz VGA controller rendering ASCII text from a BRAM font ROM, a UART receiver (19,200 baud, 8-O-1) with parity checking, and an RC4 brute-force FSM decrypting 128-bit messages (24-bit key space); closed timing at 100 MHz and validated on hardware

ESP32 Embedded Systems C, bare metal

- Wrote a register-level GPIO driver, custom ISRs and GPTimer interrupts; built ADC joystick and DAC audio drivers, Tetris with real-time TFT rendering, and ESP-NOW peer-to-peer Tic-Tac-Toe using state machines

TECHNICAL SKILLS

Languages: C, C++, SystemVerilog, Python, MATLAB, Java, SQL

Platforms: Xilinx PYNQ-Z2 (embedded Linux), Basys3 FPGA, ESP32, ATmega

Hardware & Protocols: GPIO, PWM, ADC/DAC, timers, interrupts, UART, AXI, DMA, kernel and user-space drivers

Tools: Vivado, ngspice, Qucs-S, KiCad, Git, Linux, oscilloscope, function generator, multimeter, soldering, SolidWorks (Certified)

Languages (spoken): English, Spanish, Portuguese, Japanese